

- (a) Rewrite the expression $\frac{1}{x+1} - \frac{1}{x} = \frac{1}{10}$ in the form $ax^2 + bx + c = 0$ and determine the values of a , b , and c .

$$\frac{x - (x+1)}{x(x+1)} = \frac{1}{10}$$

$$\Rightarrow \frac{-1}{x^2+x} = \frac{1}{10} \Rightarrow x^2+x = -10$$

$$\Rightarrow x^2+x+10=0$$

$$a \quad \underline{\quad 1 \quad}$$

$$b \quad \underline{\quad 1 \quad}$$

$$c \quad \underline{\quad 10 \quad}$$

- (b) Find the derivative of $A \cos(\omega t)$ with respect to t

Note: $\frac{d}{dx} \cos(x) = -\sin(x)$

$\frac{d}{dx} \sin(x) = \cos(x)$

Chain's rule: $\frac{d}{dx} f(Ax) = A \cdot \frac{d}{dx} f(x)$

$-Aw \sin(\omega t)$

- (c) Find the integral $\int_0^\infty e^{-t/\tau} dt$

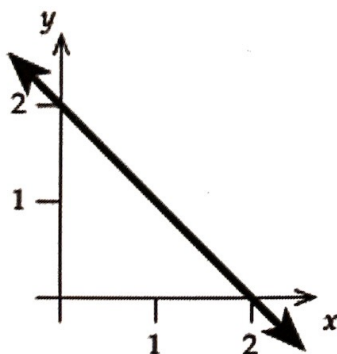
Assume $\tau > 0$.

Let $x = -t/\tau \Rightarrow t = -\tau x$

$dt = -\tau dx$

$$\int_0^\infty e^{-t/\tau} dt = \int_0^{-\infty} e^x (-\tau) dx = -\tau \int_0^{-\infty} e^x dx = -\tau \cdot \frac{1}{\tau} e^x \Big|_0^{-\infty} = -\tau(0-1) = \tau$$

- (d) Write the function that has the following graph



$y = ax + b$

Method 1: Plug in $(0, 2)$ and $(2, 0)$

Solve a, b

Method 2: Slope $a = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-2}{2-0} = -1$

Intercept $b = 2$

$y = -x + 2$

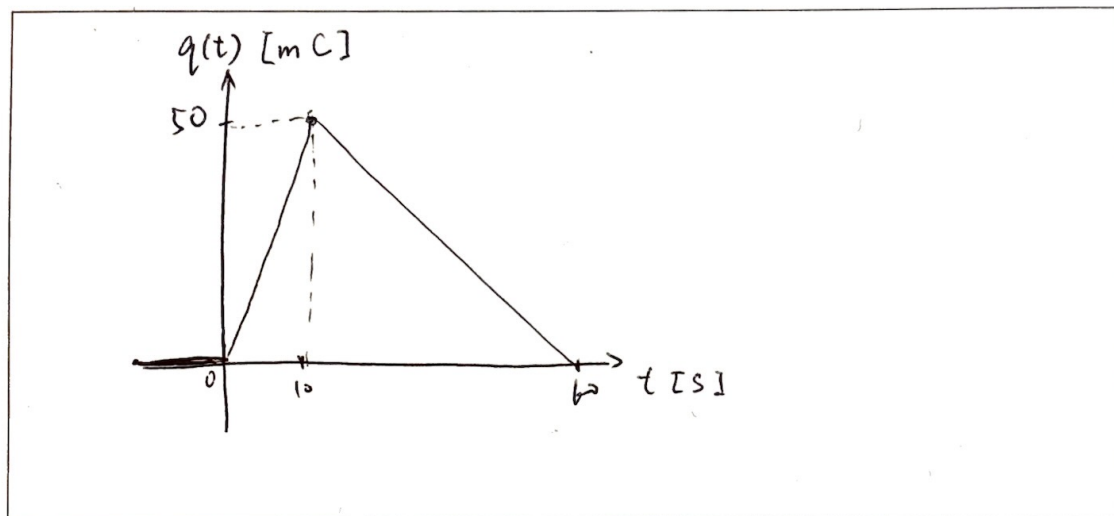
- (e) Convert $0.001 [A]$ to $[mA]$

$$10^{-3} [A] = 10^{-3} \cdot 10^3 [mA] = 1 [mA]$$

The cumulative charge in mC that entered a certain device is given by

$$q(t) = \begin{cases} 0 & t < 0 \\ 5t & 0 \leq t \leq 10[s] \\ 60 - t & 10 \leq t \leq 60[s] \end{cases}$$

Plot $q(t)$ versus t from $t = 0$ to $t = 60$ [s].

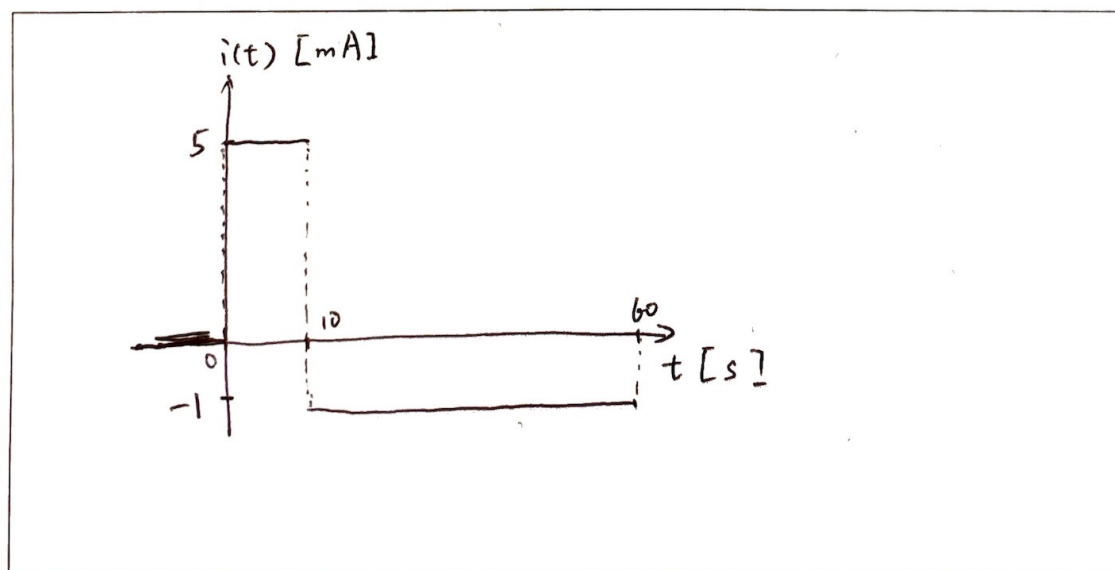


Please read the question carefully.

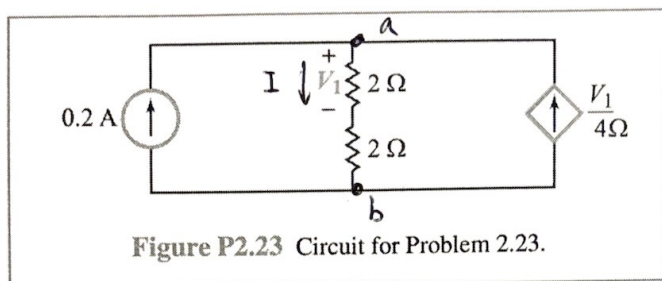
Note: $i(t) = \frac{dq}{dt}$

$$q(t) = \int i(t) dt$$

Plot the corresponding current $i(t)$ entering the device



*2.23 Determine the amount of power supplied by the independent current source in the circuit of Fig. P2.23.



KCL at node a:

$$0.2 \text{ [A]} + \frac{V_1}{4 \text{ [}\Omega\text{]}} = I$$

current entering node current exiting node

$$V_1 = \underline{0.8 \text{ [V]}}$$

$$\text{Power} = \underline{0.32 \text{ [W]}}$$

Ohm's law:

$$V_1 = I \cdot R = I \cdot 2 \text{ [}\Omega\text{]} = \left(0.2 \text{ A} + \frac{V_1}{4 \text{ [}\Omega\text{]}}\right) \cdot 2 \text{ [}\Omega\text{]} = 0.4 \text{ [V]} + \frac{V_1}{2}$$

Solve equation, $V_1 = 0.8 \text{ [V]}$

Power of independent source $P = IV = 0.2 \text{ [A]} \times V_{ab}$

2 ways of $V_{ab} = 2V_1 = 1.6 \text{ [V]}$

thinking: *. Voltage on both $2 \text{ [}\Omega\text{]}$ resistor should be identical.

Voltage sums up together when connected in series.

②. Calculate I : $I = 0.2 \text{ [A]} + \frac{0.8 \text{ [V]}}{4 \text{ [}\Omega\text{]}} = 0.4 \text{ [A]}$

$$R_{\text{total}} = 2 \text{ [}\Omega\text{]} + 2 \text{ [}\Omega\text{]} = 4 \text{ [}\Omega\text{]}$$

$$V_{ab} = I \cdot R_{\text{total}} = 0.4 \text{ [A]} \cdot 4 \text{ [}\Omega\text{]} = 1.6 \text{ [V]}$$

$$P = IV = 0.2 \text{ [A]} \times 1.6 \text{ [V]} = 0.32 \text{ [W]}$$

*. Many students answered "0.16 [W]". Please differentiate power of "source" vs. power of "resistor"